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Urban land use and land cover classification with interpretable machine learning – A case study using Sentinel-2 and auxiliary data[☆]

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ABSTRACT

The European Commission launch of the twin Sentinel-2 satellites provides new opportunities for land use and land cover (LULC) classification because of the ready availability of their data and their enhanced spatial, temporal and spectral resolutions. The rapid development of machine learning over the past decade led to data-driven models being at the forefront of high accuracy predictions of the physical world. However, the contribution of the driving variables behind these predictions cannot be explained beyond generalized metrics of overall performance. Here, we compared the performance of three shallow learners (support vector machines, random forest, and extreme gradient boosting) as well as two deep learners (a convolutional neural network and a residual network with 50 layers) in and around the city of Malmö in southern Sweden. Our complete analysis suite involved 141 input features, 85 scenarios, and 8 LULC classes. We explored the interpretability of the five learners using Shapley additive explanations to better understand feature importance at the level of individual LULC classes. The purpose of class-level feature importance was to identify the most parsimonious combination of features that could reasonably map a particular class and enhance overall map accuracy. We showed that not only do overall accuracies increase from shallow (mean = 84.64%) to deep learners (mean = 92.63%) but that the number of explanatory variables required to obtain maximum accuracy decreases along the same gradient. Furthermore, we demonstrated that class-level importance metrics can be successfully identified using Shapley additive explanations in both shallow and deep learners, which allows for a more detailed understanding of variable importance. We show that for certain LULC classes there is a convergence of variable importance across all the algorithms, which helps explain model predictions and aid the selection of more parsimonious models. The use of class-level feature importance metrics is still new in LULC classification, and this study provides important insight into the potential of more nuanced importance metrics.

[☆] All ethical practices have been followed in relation to the development, writing, and publication of the article.

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1. Introduction

Land use and land cover (LULC) classification of satellite imagery into thematic maps has been around since the launch of the Landsat program in 1972. Statistical classifiers, such as maximum likelihood (Edgeworth 1908), Mahalanobis distance (Mahalanobis 1936), or parallelepiped (Shlien and Goodenough 1974) were among the first to be used to classify remotely sensed data. These early classifiers were often parametric and based on the statistical theory that land surface reflectance should be normally distributed. Thus, they were popular due to their simplicity and the fact that remote sensing software such as ERDAS Imagine (Leica Geosystems 2004) were prepackaged with them.

As satellite data became more readily available and multitemporal analysis more viable, statistical classifiers were replaced by more sophisticated algorithms. These complex algorithms are based on a myriad of different architectures including hyperplanes (e.g. support vector machines (Cortes and Vapnik 1995)), decision trees (e.g. random forests (Breiman 2001)), and neural networks (e.g. deep learning (Bengio 2009)). The shift from statistical classification to machine learning in LULC classification stemmed largely from the fact that statistical classifiers are reliant on the underlying statistical properties of the data. For example, maximum likelihood assumes that the data is normally distributed, Mahalanobis distance assumes that covariances of all the classes are equal, and parallelepiped uses the means and standard deviations of each class to construct patterns. These assumptions about the data create limitations that reduce, in relative terms, the accuracies of these classifiers (Hansen et al., 1996; Olson 2009; Richards 2005).

Both support vector machines (SVM) and random forests (RF) were developed in the 1990s, just prior to an era of rapid advancements in computer chip multithreading that began in the 2000s (Barroso 2005; Spracklen and Abraham, 2005). This convergence of technological development meant that both algorithms were rapidly adopted in remote sensing science, particularly for LULC classification (Pal 2005; Pal and Mather 2005). Recent advancements in machine learning have involved solving optimization problems in order to enhance scalability of algorithms so that they can efficiently handle Big Data. A particular algorithm that has found an increasing number of users in the remote sensing community is extreme gradient boosting (XGB) (Chen and Guestrin 2016), which is an optimized variant of gradient boosting machines (Friedman 2001). Another family of algorithms that has witnessed rapid progress over the past two decades has been artificial neural networks (ANNs). ANNs have been used in remote sensing since at least the 1990s to classify satellite imagery with generally modest classification accuracies that did not particularly outperform non-ANN algorithms (Wilkinson 2005). Progress in computing capacity and Big Data post-2010 has created an ecosystem ripe for more sophisticated ANN architectures that are collectively called deep learning (DL) (Goodfellow et al., 2016; LeCun et al., 2015). These have renewed interest in ANNs due to their high classification accuracies (Ma et al., 2019) and the availability of models pretrained on deeper architectures (He et al., 2016). Depending on the application, DL algorithms have shown to outperform RF, SVM or XGB in a variety of cases involving satellite image classification (Kussul et al., 2017; Ma et al., 2017; Zhong et al., 2019) but their adoption in remote sensing relative to those algorithms has been subdued due to, among other things, hardware requirements (e.g. graphical processing units) for tasks involving satellite imagery (Ball et al., 2017).

The increased availability of satellite data in the 21st century meant that there has been a parallel increase in the diversity of algorithms employed to classify these data and extract information from them. This spurred several studies (Abdi 2020; Jozdani et al., 2019; Maxwell et al., 2018; Talukdar et al., 2020) to compare classifiers to shed light on their performance in different scenarios. Similarly, as auxiliary data became more readily available, research also focused on finding the optimal number of input variables in order to reduce noise (Shao et al., 2016) and remove redundant variables (Belgiu and Drăguț 2016). The integration of auxiliary data to improve the classification accuracy of remote sensing data has been known for at least 40 years (Hutchinson 1982). Auxiliary data can complement spectral data by providing additional information about, for example, topography (Hurskainen et al., 2019), land surface phenology (Xue et al., 2014) or the spatial configuration of the landscape (Rodríguez-Galiano et al., 2012).

Though auxiliary variables can help improve classification accuracy, the relative importance of these and the spectral data is dependent on the complexity, components and configuration of the landscape. The classifier used also influences variable importance because, depending on algorithm architecture and aggregation scheme, certain variables are given more weight than others (Sage et al., 2020). In random forests, for instance, variable importance is based on the mean decrease in accuracy (MDA) when a particular variable is removed from a model (Breiman 2001). Thus, for the out-of-bag sample of each tree in the forest, the algorithm calculates its overall accuracy and then sequentially permutes the variables to derive the decrease in accuracy without the variable in question. Thus, the larger the MDA induced by the removal of a variable, the higher its importance in the variable hierarchy.

The identification of variables that are important for LULC mapping can help streamline the classification process by focusing on the key variables necessary for the desired outcome. For example, variable importance metrics have been used to optimize the feature space by selecting a subset of the data without compromising accuracy (Belgiu and Drăguț 2016). Variable importance metrics also aid the interoperability of models and shed light on the role of biophysical attributes in the classification process (Corcoran et al., 2013). For instance, the shortwave infrared portion of the electromagnetic spectrum (1100–3000 nm) is sensitive to canopy water content in forested ecosystems (Lukeš et al., 2013) whereas the red edge band centered at 740 nm (Horler et al., 1983) is capable of differentiating herbaceous crops from grasses based on their chlorophyll concentrations (Radoux et al., 2016). These examples point to the utility of understanding variable importance metrics in order to construct models that are both interpretable and yield reasonably high accuracies to stimulate replication.

High accuracies, both on overall and on a per-class basis, is the desired objective of most LULC classification exercises. State-of-the-art machine learning algorithms have shown to perform well in satellite image classification, often resulting in overall accuracies that exceed the 90th percentile (Ma et al., 2019). Generally, the higher the classification accuracy and learning rate of an algorithm, the more difficult it is to interpret its results (Gunning et al., 2019). High-accuracy blackbox models have traditionally not been con-

ductive to interpretability in part because of a lack of clarity regarding the mechanism through which the class-wise predictions are made.

Depending on the application, model interpretability may not be a priority for certain end-users. Consider, for example, an investor interested in using satellite imagery to detect and count the number of cars at the parking lot of a shopping center to decide whether the center would make a profitable investment. In this case, the end-user is more interested in a very high predictive accuracy than the mechanisms behind the predictions in order to make a decision. Nevertheless, additional information about how the model computed the prediction can be beneficial to the end-user. In the example of the shopping center, it could be that temporal information (such as the day of the week), and spatial attributes of the location (distance to business and residential districts or to highways) rank high as important variables and this could be additional information that could help the investor to make a decision. Thus, a model with an interpretable mechanism can increase confidence among stakeholders such as businesses, communities, land managers and policymakers of the classification it produces (Campos-Taberner et al., 2020). This is perhaps the reason behind why some national mapping programs resort to easily interpretable algorithms such as maximum likelihood in the case of Sweden (Naturvårdsverket 2018) or decision trees in the United States (Jin et al., 2019).

Several machine learning algorithms are currently implemented in LULC mapping and many more are making the transition from the computer/data science domain to remote sensing. As these algorithms are often based on disparate underlying architectures, the variable importance hierarchies they produce may not be the same. That said, generalizable consistencies in variable ranking can point to coherent subsets of variables that prove crucial for classifying the LULC classes in question. Thus, the overarching objective of this study is to explore the interpretability of popular machine learning algorithms using Shapley additive explanations (Lundberg and Lee 2017; Shapley 1953) to better understand feature importance at the level of individual LULC classes. The intention here is to understand which variables consistently stand out across classifiers in order to make biophysical inferences. We use a suite of geospatial data, both spectral and auxiliary, in a complex, mixed-use, urban and rural landscape to better understand the importance of variables in LULC classification using five popular algorithms. We ask two main questions: (1) Which input variables are important for LULC classification, and what spatial or spectral characteristics set them apart? (2) Is there convergence in the importance of input variables across the five algorithms?

2. Methods

2.1. Study area

The study area is Malmö municipality (55°34'51"N, 12°59'59"E) and its immediate surroundings (Fig. 1). Malmö is the third-largest urban area in Sweden with a population of nearly 350,000 (Statistics Sweden 2021) and covers a land area of 160 km². The climate is temperate with a mean annual temperature and rainfall of 10 °C and 600 mm, respectively (SMHI et al., 2021). The topography has a low relief with only a 73 m change in elevation between sea level and the highest point on the land surface. The urban landscape of the municipality is compact yet diverse. It is bordered by the Öresund strait to the north and east and features a densely configured urban center and an adjacent harbor and industrial zone. The landscape gradually opens to a lower density residential area in the south and west, eventually ending in an agricultural landscape. The agricultural land around these margins is primarily composed of maize, wheat, rapeseed, and pasture (SBA 2018). The city has been experiencing a rise in housing demand over the past three decades and has green space coverage of around 46% (Olsson et al., 2020). Although there are few areas with dense tree canopy cover, trees across the city are primarily deciduous and dominated by *Sorbus*, *Tilia*, *Acer*, *Purnus*, and *Fagus* species (Sjöman et al., 2012).

2.2. Data

2.2.1. Sentinel-2 data

The Sentinel-2 Multispectral Instrument (MSI) is composed of two satellites (Sentinel-2A and Sentinel-2B) that provide data in three spatial resolutions: 10m, 20m, and 60m (Drusch et al., 2012). We downloaded MSI data covering tile 33UUB in 2017. The criteria for selecting the satellite imagery were that the MSI scenes must not contain haze or clouds and that they must cover different seasons to capture vegetation phenology. A total of four scenes (January 27, March 11, April 30, and May 27) satisfied our criteria and were further processed. The scenes were downloaded from the Copernicus Open Access Hub as Level-1C top-of-atmosphere products that have undergone geometric correction, including spatial registration and orthorectification, with sub-pixel accuracy but that require atmospheric correction (ESA 2015). However, certain Sentinel-2A Level-1C scenes have been shown to contain band-to-band registration (BBR) errors (Yan et al., 2018) that can cause spectral distortions. The multitemporal BBR was assessed qualitatively by inspecting the temporal profile of each band for pixels at interface of two contrasting LULCs. We then used *Sen2Cor* (v2.5.5) to correct for atmospheric effects and convert the data to Level-2A land surface reflectance (Müller-Wilm 2018). The 60 m bands were not used in this analysis and the 20 m bands were up-sampled to 10 m using bilinear interpolation. Each of the resultant 10 bands was named as follows: B, followed by the band number, and a three-letter code for the month. For example, Band 4 (the red band centered at 665 nm) for the month of March would be abbreviated as B4MAR.

2.2.2. Land surface phenology

The large-scale detection of seasonal vegetation dynamics from space is termed land surface phenology (LSP) and has been found to be a useful predictor in classifying satellite imagery (Jia et al., 2014; Nguyen and Henebry 2019; Qader et al., 2016). We used the LSP data for 2017 that were generated in a previous study (Abdi et al. (2021); Supplementary Figure S1) as input into our classification workflow. The method implemented in producing the LSP data was based on the approach of Jönsson et al. (2018). Briefly, it in-

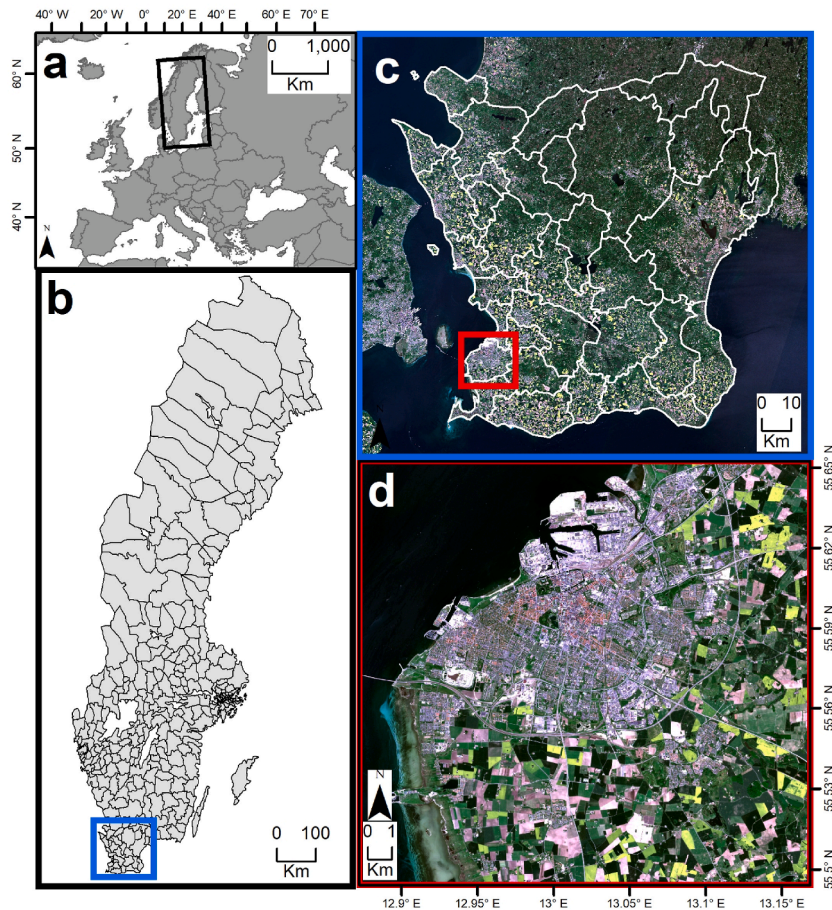


Fig. 1. Overview of Malmö municipality relative to Europe (a), Sweden (b), and the province of Skåne (c). The urban fabric of Malmö (d) comprises of a high-intensity inner portion, a low-intensity outer portion and is surrounded by agricultural land.

involves the reconstruction of daily Sentinel-2 time series using a box constrained seasonal shape prior that is fit to a double logistic function. Readers interested in a detailed mathematical description of the approach are encouraged to consult [Jönsson et al. \(2018\)](#).

2.2.3. Topography

We downloaded a digital elevation model (DEM) provided by Copernicus Land Services (EU-DEM v1.1) for tile E40N30. The DEM was provided at a spatial resolution of 25 m and resampled to 10 m using bilinear interpolation. Then, the slope and aspect were extracted as additional topographic variables for inclusion into the classification workflow. Topographic variables were included in order to help detect changes in LULC linked to topography such as near-shore land reclamation.

2.2.4. Vegetation indices

We calculated three widely used vegetation indices from the Sentinel-2 spectral bands – normalized difference vegetation index (NDVI, [Rouse et al. \(1974\)](#)), two-band enhanced vegetation index (EVI2, [Jiang et al. \(2008\)](#)), and normalized difference red edge index (NDRE, [Barnes et al. \(2000\)](#)). These indices were included as they are linked to vegetation chlorophyll content and have been used to estimate vegetation growth, maturity and biomass ([Xue and Su 2017](#)). They were included to further aid the separation among vegetation classes in the study area.

2.2.5. Satellite image texture

Image texture refers to the heterogeneity in the tones within a defined area of an image. Knowledge about extracting textural information from satellite imagery has been known since the launch of the first Landsat satellite ([Haralick et al., 1973](#)). Since then, satellite image texture was used to provide information on vegetation structure ([Wood et al., 2012](#)), urban population density ([Liu et al., 2006](#)) and land use ([Herold et al., 2003](#)). Within a 3×3 kernel, we implement gray-level co-occurrence matrix (GLCM) and compute second-order image texture that is rotationally invariant, i.e. average texture using shift values of 0° , 45° , 90° , and 135° in order to consider the relationships between adjacent pixels thereby accounting for spatial configuration of the landscape ([Haralick 1979](#)). The 3×3 window was chosen specifically because the study area is fairly small and there is considerable structural diversity within it ([Fig. 1d](#)), furthermore we want to capture local-scale variation within the urban fabric. The 10 m resolution of Sentinel-2 bands

yields a 900 m² area within the window that is sufficient to extract small changes (Zhang et al., 2001). Two GLCM features were computed for each Sentinel-2 band – *Correlation* and *Entropy* – resulting in a total of GLCM 80 layers. These two features were selected for two reasons; (1) Most of the texture measures are strongly correlated with each other, for example *Entropy* is strongly correlated with *Angular Second Moment* ($r^2 = -0.87$) so we chose one of them, and (2) *Correlation* is more independent as it shows a correlation of less than 0.5 with any other GLCM feature (Hall-Beyer 2007, 2017).

2.2.6. Reference data

Reference data was downloaded from the Swedish Mapping, Cadastral and Land Registration Authority (*Lantmäteriet*) and the Swedish National Land Cover Database (*Nationella Marktäckedata*). These data included georeferenced (EPSG:3006) polygons on water bodies, wetlands, agricultural areas, urban areas (high intensity, low, intensity, industrial), and recreational areas including parks, beaches, sports facilities, shopping centers, and the like. These features were superimposed on high-resolution imagery from Google Earth and rearranged into a total of eight LULC classes were defined for this study (Fig. 2): (1) Agriculture, (2) Deciduous Trees, (3) Non-vegetated Open Land, (4) Water, (5) Wetland, (6) High Intensity Urban, (7) Low Intensity Urban, (8) Industrial. The rearrangement involved the combination of certain classes and the disaggregation of other classes that were provided by *Lantmäteriet*. For example, we disaggregated urban areas into Low Intensity, High Intensity and Industrial, and integrated recreational areas into the Low Intensity Urban.

2.3. Classification approach

2.3.1. Training, validation and testing

We randomly selected 1000 reference training samples (see section 2.2.6) for each class using stratified sampling. For validation, we selected 200 stratified random samples of each class from the reference samples for input into the hyperparameter optimization. Finally, 10,000 samples per class were selected from the remaining reference samples for testing the model performance (Table 1).



Fig. 2. Close-up of the eight land use and land cover classes found within the study area. The High and Low Intensity Urban differ primarily in the amount of impervious surfaces, green spaces, single family homes, and multi-family apartment complexes. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 1

Distribution of the reference samples for model training, validation, and testing.

Class	Training	Validation	Test
(1) Agriculture	1000	200	10000
(2) Deciduous Trees	1000	200	10000
(3) Non-vegetated Open Land	1000	200	4523
(4) Water	1000	200	10000
(5) Wetland	1000	200	10000
(6) High Intensity Urban	1000	200	10000
(7) Low Intensity Urban	1000	200	10000
(8) Industrial	1000	200	10000
Total	8000	1600	75523

The Non-vegetated Open Land class had 4523 test samples due to the small areal coverage of this class. To obtain more robust results, each classifier is trained through a five-fold cross-validation and the averaged results are reported as the main performance of the model. A flowchart showing key steps is shown in Fig. 3.

2.3.2. Shallow machine learning methods

This section provides a brief overview of the shallow machine learning (ML) methods in this study. Readers seeking detailed descriptions are encouraged to review the literature cited at the end of each algorithm subsection.

2.3.2.1. Random forest. Random forest (RF) is an ensemble classifier of several bootstrap aggregated decision trees (Breiman 2001). Basically, RF employs bootstrap aggregation to reduce the variance in decision trees and then estimates their parameters by choosing a random subset of training samples, subject to minimizing the correlation with other trees. Each decision tree results in an output that collated and the final result is produced via an unweighted mean majority vote of all the trees. More details of the RF algorithm are provided in Breiman (2001).

2.3.2.2. Extreme gradient boosting. Extreme gradient boosting (XGB) is an optimization of gradient boosting machines. It comprises shallow decision trees as weak classifiers. These classifiers are optimized based on an objective function, and in order to avoid overfitting, their complexity is controlled by a regularization term. We used cross entropy loss in this study, which is a standard loss function for multi-class LULC classification. Chen and Guestrin (2016) provide a detailed description of XGB algorithm.

2.3.2.3. Support vector machine. Support vector machine (SVM) models the input data by mapping them into higher dimensions using linear or nonlinear kernel functions and then fits a hyperplane based on the training samples. SVM attempts to maximize the margin between the estimated hyperplanes by solving an optimization problem. In other words, the algorithm tries to find samples from each class in a hyper-dimensional space and fits a linear hyper-plane in a way that the defined distances between two planes are maximal. The chosen samples are known as support vectors. It is worth mentioning that SVM is a binary classifier. There are two main approaches to extending SVM to a multi-class classifier: one-versus-rest (OVR) and one-versus-one (OVO). We used the OVR to resolve multi-class LULC classification. A detailed description of SVM is provided in Cortes and Vapnik (1995).

2.3.2.4. Model training and hyperparameter optimization. ML algorithms comprise hyperparameters that need to be adjusted (i.e. tuned) in order to obtain optimum results. In this study, hyperparameters were tuned during the training stage based on the custom random grid-search approach. Accordingly, 1000 and 200 random samples of each class were selected from the reference data for generating training and validation data, respectively. In a conventional grid search algorithm (Bhat et al., 2018), each ML model is trained for all combinations of the hyperparameters in the grid, and the hyperparameters that produce the best result are selected.

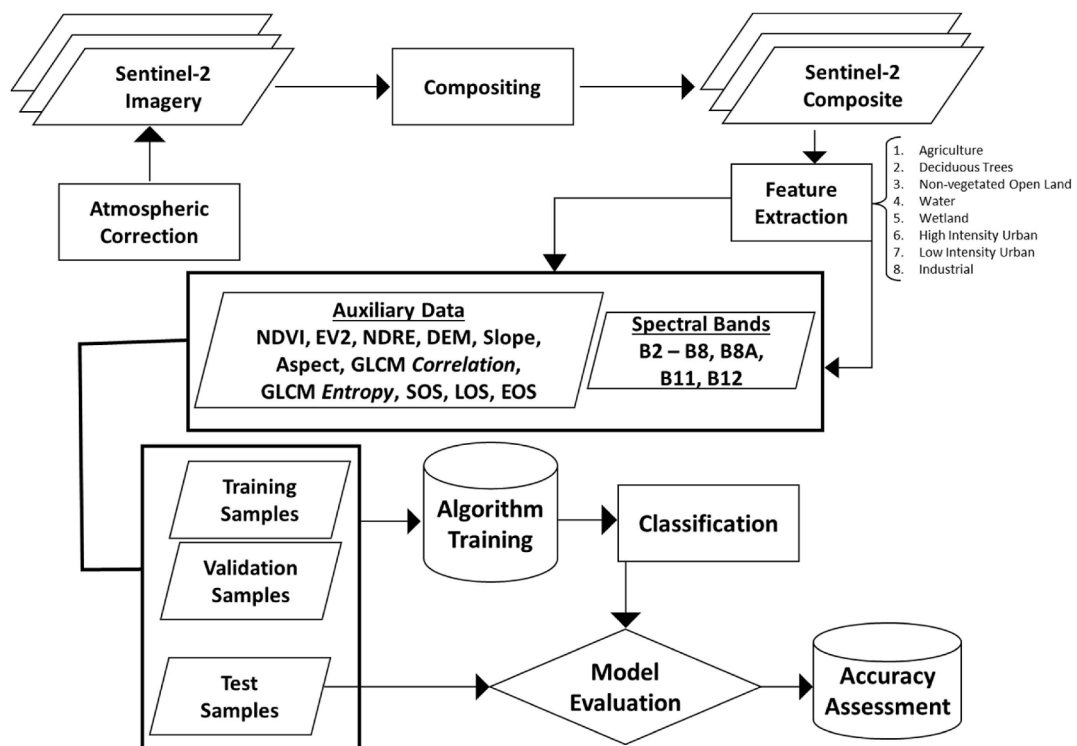


Fig. 3. Flowchart of the methodological approach.

In our customization, this process is repeated in multiple iterations and at the end of each iteration the search space becomes smaller. This enables the optimization algorithm to go deeper in the search space and find a better optimum point. Therefore, each ML model was optimized based on the proposed algorithm in 10 iterations, whereby at each iteration the hyperparameter values were randomly sampled between their maximum and minimum bounds. Details of the optimization process are shown in Table 2.

2.3.3. Deep machine learning methods

Deep machine learning (DL) simultaneously uses graph technology and data transformations within a multilayer feature-learning architecture (Alzubaidi et al., 2021; Zhu et al., 2017). There are a variety of DL architectures, including recurrent neural network (RNN), feed-forward multi-layer perceptron (MLP), and convolutional neural network (CNN) (Goodfellow et al., 2016). CNNs have shown better performance in image processing and classification than other architectures because their local connection and shared weights within the algorithm make full use of the inherent 2D structure (i.e. spatial and spectral dimensions) of satellite data (Alzubaidi et al., 2021).

This study employs two types of patch-based CNN architectures: a simple feed-forward 2D CNN and a residual network (ResNet) (He et al., 2016) representing a more complex model framework. In particular, residual modules facilitate the training process and overcome the vanishing gradient problem in the backpropagation process of very deep models by adding a direct path from input to output. Therefore, the deep model can be trained more efficiently while not adding any extra computational complexity. Accordingly, this model architecture has demonstrated promising results in many kinds of remote sensing community research in the past few years, making it more special to investigate. We implemented a ResNet with 50 layers (ResNet50) as it was shown to perform well in LULC classification (Scott et al., 2017; Tong et al., 2018). A detailed description of the ResNet can be found in He et al. (2016).

Notably, two common approaches exist for training a DL model: transfer learning (TL) and training the model *ab initio*, i.e. letting the model learn from the input data itself without any prior learning. TL uses weights of models that were trained using extensive datasets such as ImageNet (Deng et al., 2009), which comprises 1.2 million photographic images (Scott et al., 2017). In contrast, model training *ab initio* initializes with random or zero-valued weights without pre-trained weights from other models. An advantage of TL is that it prevents the model from overfitting when limited training data is available (Hu et al., 2015) and training starts with pre-trained weights, thus the model can converge with fewer epochs (Karim and van Zyl 2021). However, input data size needs to be the same as the pre-trained model (e.g., $224 \times 224 \times 3$). This somewhat limits the applicability of TL for LULC using satellite imagery because almost most of available pre-trained models have been trained on RGB datasets, and as a result are not optimized on other multi-channel imagery.

We used Sentinel-2 multispectral bands and auxiliary feature layers using an input tensor size of $(31 \times 31 \times b)$. Given that each ground pixel has been resampled as 10 m in height and width, each input patch covers an area of 310^2 square meters, and b is the number of channels that varies according to each investigated scenario. Thus, the dataset we used to train the algorithms consists of more than three RGB layers and as a result we train our deep models *ab initio* and initialize the model weights with random values. Table 3 summarizes the hyperparameter implementation of the two DL models in this study. The considered simple CNN model consists of three 2D convolutional layers followed by 2×2 Max pooling operators. In contrast, ResNet50 consists of one simple convolution and Max pooling layer and four residual blocks. Both of these deep networks are connected to a Softmax fully connected layer as a classifier, and the model is trained by the backpropagation of categorical cross-entropy loss value by Adaptive momentum optimizer (Kingma and Ba 2014).

2.3.4. SHapley Additive exPlanations (SHAP)

Complex models obtained by ML and DL methods are perceived as black-box models, where the link between input data and the model prediction boundaries is blurred and hard to interpret. However, model interpretation is essential to understand the relationship between the data and the model if the latter is to be used in real-world conditions.

Shapley additive explanations (SHAP) (Lundberg and Lee 2017; Shapley 1953) is a method to explain the contribution of input data layers or features to the final predictions based on the Shapley values. According to game theory, in a coalition game, each agent (player) contributes to the final result. Shapley value is used to determine the effectiveness of each agent, and it is the only solution for fair payment to an agent based on its contribution and usefulness to the game (Shapley 1953). In particular, Shapley value is the marginal contribution of an agent among all the possible combinations, and it can be computed by Equation (1):

Table 2

List of hyperparameter grid settings to obtain optimum values for machine learning models.

Model	Search space
SVM	Kernels = 'RBF', 'Sigmoid' C = 10^{-10} ; 10^{-10} Gamma = 10^{-10} ; 10^{-10}
XGB	Lr = 10^{-5} ; 10^{-1} n_estimators = 200:2000 max_depth = 0:100
RF	n_estimators = 200:2000 max_depth = 0:100

Table 3

Details of the hyperparameters implemented in the deep learning models.

Model	Layer type	Kernel size	Stride	Filters	repeats
CNN	Simple convolution	3×3	1	32	1
	Max pooling	2×2	1	–	1
	Simple convolution	3×3	1	64	1
	Max pooling	2×2	1	–	1
	Simple convolution	3×3	1	128	1
	Max pooling	2×2	1	–	1
ResNet50	Simple convolution	7×7	2	64	1
	Max pooling	3×3	2	–	1
	Residual	1×1	1	64	3
		3×3		64	
		1×1		256	
	Residual	1×1	1	128	4
		3×3		128	
		1×1		512	
	Residual	1×1	1	256	6
		3×3		256	
		1×1		1024	
	Residual	1×1	1	512	3
		3×3		512	
		1×1		2048	

$$\varphi_i = \frac{1}{|F|!} \sum_{S \subseteq F \setminus \{i\}} |S|! (|F| - |S| - 1)! [f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_S(x_S)] \quad (1)$$

where, $f_{S \cup \{i\}}$ and f_S are the model with and without contribution of the i th agent, thus the term $[f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_S(x_S)]$ calculates the marginal contribution of the i th agent to the game when it is added. This process should be repeated for all possible subsets of $S \subseteq F$ within F agents. Accordingly, $|S|!$ is the possible way to form subset S prior to adding the i th player, and $(|F| - |S| - 1)!$ is the amount of ways that the remaining agents could be added.

SHAP provides a framework for estimating feature contributions in an ML/DL model by adapting them as a coalition game and by estimating a feature's Shapley values. In general, each input feature can be considered as an agent, while the model predictions (outputs) are equivalent to the payouts. In particular, the SHAP algorithm utilizes additive feature attribution (Lundberg and Lee 2017) in order to approximate and explain the complex models. The original prediction model $f(x)$ with the input of x is approximated by the simplified explanation model $g(x')$ with the simplified input of x' . This approximation is accomplished based on the linear function of binary variables, as shown in Equation (2):

$$g(x') = \varphi_0 + \sum_{i=1}^M \varphi_i x'_i \quad (2)$$

where $x'_i \in \{0, 1\}$ is a binary-formatted simplified variable of x , M is the number of simplified input features, which explains the availability (one) or unavailability (zero), φ_0 is the empty set attribution that is mean of the complete dataset predictions (Lundberg and Lee 2017). The contribution of each feature based on Shapley values (φ_i) (Eq. (1)) becomes part of a linear function of simplified binary variables. Therefore, according to equation (2), it is possible to approximate the $f(x)$ based on the additive feature attribution explanation model. Five standard quantitative metrics were used for evaluating the model outputs. These include Precision (Pr) and Recall (Re), F1 score, Overall Accuracy (OA), Kappa coefficient (k) and Average Accuracy (AA).

2.3.5. Experiments

We considered 17 scenarios (Table 4) in order to understand the variations model accuracy contributed by different categories of input variables (i.e. Sentinel-2 multispectral data, land surface phenology, topography, vegetation indices, and GLCM features). All 17 scenarios were included as input data for evaluating the five ML/DL algorithms (RF, XGB, SVM, CNN, and ResNet50). Finally, based on this combination of input datasets and classifiers, a total of 85 scenarios were generated and evaluated. Table 4 summarizes all the scenarios for input layers. The number of input layers for each scenario varies between 40 layers, when only Sentinel-2 data are used, up to 141 layers that include the full suite of input variables.

Model intercomparison was performed using a two-proportion Z-test with Yates' continuity correction that compares the proportion of correctly classified (PCC) pixels of two algorithms. The test outputs the probability of whether the PCC of the algorithm pair is not different from one another (null hypothesis). Additionally, McNemar's chi-squared (χ^2) test (McNemar 1947) was used to test for the class-wise marginal homogeneity by comparing the error matrices of two algorithms at a time. The purpose of this method is to statistically show whether predictions on a per-class basis for each algorithm pair displays statistically insignificant difference (i.e. they are equal) by comparing the algorithm predictions to each other.

Table 4

The 17 classification scenarios and their corresponding input layers.

Scenario no.	Sentinel-2	Phenology	Topography	Vegetation Indices	GLCM	Number of Layers
1	✓	-	-	-	-	40
2	✓	✓	-	-	-	46
3	✓	-	✓	-	-	43
4	✓	-	-	✓	-	52
5	✓	✓	✓	-	-	49
6	✓	✓	-	✓	-	58
7	✓	-	✓	✓	-	55
8	✓	✓	✓	✓	-	61
9	-	-	-	-	✓	80
10	✓	-	-	-	✓	120
11	✓	✓	-	-	✓	126
12	✓	-	✓	-	✓	123
13	✓	-	-	✓	✓	132
14	✓	✓	✓	-	✓	129
15	✓	✓	-	✓	✓	138
16	✓	-	✓	✓	✓	135
17	✓	✓	✓	✓	✓	141

SVM, RF, and XGB were run using Scikit-learn package (Pedregosa et al., 2011), CNN and ResNet50 were trained using the Tensorflow package (Abadi et al., 2016), and the grid search algorithm was implemented in Python 3 (Van Rossum and Drake 2009). Visualization of the results was performed in ArcGIS 10.5 (ESRI 2011).

3. Results

3.1. Algorithm performance

Across all the tested algorithms, the mean overall accuracy (OA) of the best performing scenario was $87.8 \pm 4.5\%$ (Table 5). Within these scenarios, the highest OA (92.95%) was produced by the ResNet50 model with only the Sentinel-2 multispectral data as input variables and RF had the lowest OA (83.16%) with the complete suite of auxiliary variables (Sentinel-2 multispectral, Phenology, Topography, Vegetation, GLCM). There was considerable variation in the 17 scenarios for all the classifiers (Supplementary Tables S1 – S5) as the inclusion of additional variables did not necessarily result in higher OA. For example, algorithms based on decision trees, such as RF and XGB, the inclusion of additional variables (except Phenology) to the Sentinel-2 baseline increased OA by an average of 0.85% and 1.87%, respectively. In the case of SVM, there was a general decline in OA as the number of variables increased. However, this depended on the type of variable that was included. The decline in OA was more pronounced when the GLCM variables were included in the suite (Supplementary Table S3). The DL models both crossed 90% OA threshold without any of the auxiliary data. For the ResNet50 model, the Sentinel-2 multispectral data were sufficient for the algorithm to reach maximum OA (92.95%). In contrast, the CNN model reached its maximum OA (92.31%) with the inclusion of the topography variables (DEM, slope, aspect) to the Sentinel-2 multispectral data.

The radar chart of the precision and recall metrics for the individual classes in the best performing models (Fig. 4) displayed a consistent pattern whereby the DL models performed better ($> 90\%$) in the built-up classes (High- and Low Intensity Urban, and Industrial) than the ML models. RF particularly underperformed in the High Intensity Urban class despite the fact that the model included GLCM features. The precision of the ML models clearly underperformed compared to the DL models in the Non-vegetated Open Land class. However, the recall metrics did not produce a stark difference between ML and DL models for that class.

The classified maps produced from each classifier's best performing model were similar (Supplementary Figure S2), and different levels of abstraction employed by each classifier are evident in the visual details of each class (Fig. 5). For certain classes with homogeneous coverage, such as Agriculture or Water, the difference between the classifiers was minimal. More complex classes such as Wetlands and Low Intensity Urban displayed clear differences between classifiers. For example, the Low Intensity Urban class includes suburban areas that comprise houses with gardens and trees. This greenery is captured by the ML algorithms and shows up as a

Table 5

Overall results of the best performing scenarios across all five algorithms. The model with the highest overall accuracy is shaded in gray. RF = Random Forest, XGB = Extreme Gradient Boosting, SVM = Support Vector Machine, CNN = Convolutional Neural Network, ResNet = Residual Network, OA = Overall Accuracy, AA = Average Accuracy, APr = Average Precision, AF1 = Average F1.

Model	OA	Kappa	AA	APr	AF1	Best scenario
RF	83.16	80.68	83.38	82.25	82.52	Sentinel-2 multispectral, GLCM, Phenology, Topography, and Vegetation
SVM	84.68	82.41	84.56	83.57	83.91	Sentinel-2 spectral and Topography
XGB	86.09	84.03	86.10	85.26	85.58	Sentinel-2 multispectral, GLCM, Phenology, Topography, and Vegetation
CNN	92.31	91.17	92.36	91.86	92.06	Sentinel-2 multispectral and Topography
ResNet50	92.95	91.90	92.67	92.58	92.61	Sentinel-2 multispectral

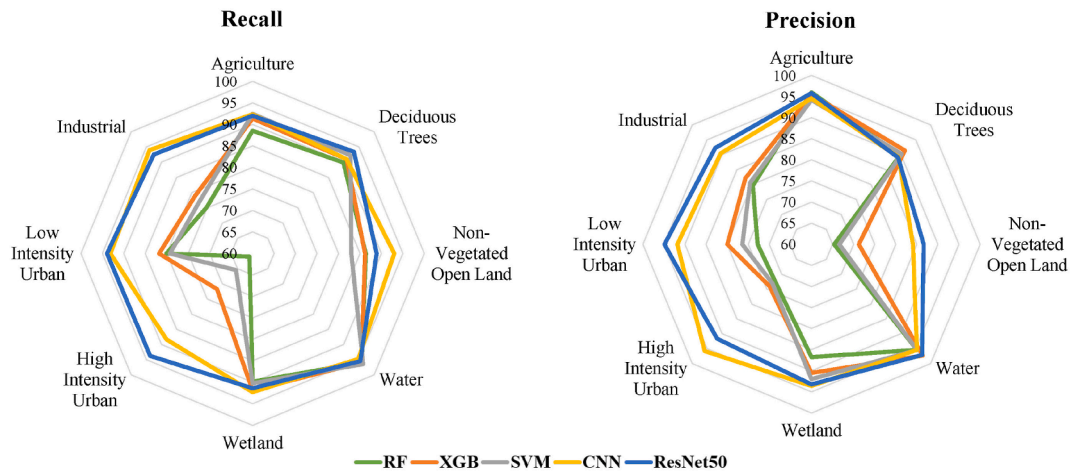


Fig. 4. Comparison of the recall and precision of the five algorithms and the eight land use and land cover classes.

speckled effect in the classified maps. In contrast, the DL models provide a fairly homogeneous representation of the classes, which mimics the spatial configuration of the training data.

The two-proportion Z-test showed that six algorithm pairings were statistically insignificant (i.e. they are similar) while four were statistically significant (i.e. they are dissimilar) (Table 6). The mean difference in the correctly classified pixels between the insignificant algorithm pairings was 2433 pixels, while between the significant pairings the mean difference was 6496 pixels. The threshold where the difference in correctly classified pixels shifts from insignificant to significant is 5400 pixels, which is the median value of the highest difference between insignificant pairings and the lowest difference between significant ones. The null hypothesis of the McNemar's test is that neither of the two models being considered is better than the other, a statistical significant indicates that the alternative hypothesis is true, i.e. one of the models is better than the other. On a class-wise bases, the McNemar's test results showed 79% of class-wise marginal homogeneity between paired algorithms showed was significant at the 1% level ($P \leq 0.01$) indicating that predictions of the algorithm pairings were statistically distinctive from one another despite the relative proximity of their overall accuracies. There were some exceptions, such as the Wetland class, whose classification accuracy was statistically insignificant (i.e. similar) across the pairings of XGB – SVM, XGB – CNN, XGB – ResNet50.

3.2. SHAP feature importance

The effectiveness and contribution of each input variable for each classifier were calculated and sorted based on the SHAP algorithm, which is the mean of the absolute amount of Shapley values. Fig. 6 depicts the importance ranking for each feature. Each row demonstrates the Shapley value that is summed for each class and variable. The higher position of the feature in the bar chart implies the higher importance and contribution. Note that only the first 20 effective features are shown in Fig. 6, but the complete list is provided in the Supplementary Information. As can be observed, these results are in line with the effectiveness of multispectral layers compared to the rest of the input variables. Furthermore, the results show the very low contribution of GLCM features in their corresponding classifiers; almost none of them were recognized within the top 20 effective features, and most of their effects were negligible compared to the multispectral, topographic, or phenological features.

According to the Shapley values, the order of importance of the input data for each class was dependent on the classifier. Some consistencies are evident – for example, B7MAY, which is one of the red edge bands centered at 783 nm, was important for the Agriculture class for all the algorithms except ResNet50, in which case B8AMAY (red edge centered at 865 nm). Another consistency is one split between ML and DL for the Water class. In this case, B12MAY (shortwave infrared at 2190 nm) was consistently important for ML classifiers and B8MAY (near infrared at 842 nm) for the DL classifiers. Some importance rankings were more predictable across classifiers. For example, the chlorophyll-sensitive B8MAY, B8AMAY, and EVI2MAY (2-band enhanced vegetation index), were all important for Deciduous Trees. Other rankings came as a surprise, for instance, the importance of DEM for High Intensity Urban and Industrial. DEM was also the second-most important auxiliary variable for the Water class (Supplementary Table S6).

Mapping the most important input variable for each class and classifier (Fig. 7) reveals the spatial pattern of variable importance. Although the algorithms generally had different variables for characterizing different urban classes, SVM exhibited a different behavior by choosing B11MAR to represent both the Low- and High Intensity Urban areas. For all classifiers, important variables for Wetland characterization varied between multispectral and topographic. Interestingly, DEM was important for two completely different classes, Wetland and High Intensity urban, in RF and XGB, respectively. Only one phenology variable, EOS, is present amongst the most-important variables and it captures the Non-vegetated Open Land in the CNN model. The High Intensity Urban and Industrial were the only two classes where each classifier displayed a different variable that was most-important. There was some overlap in the variable importance for the remaining classes.

The dominance of the Sentinel-2 multispectral data on the Shapley additive explanations is clearly shown in the skyline plot (Fig. 8). In particular, the near- and shortwave infrared layers were particularly important across all the classes. The vegetation indices and

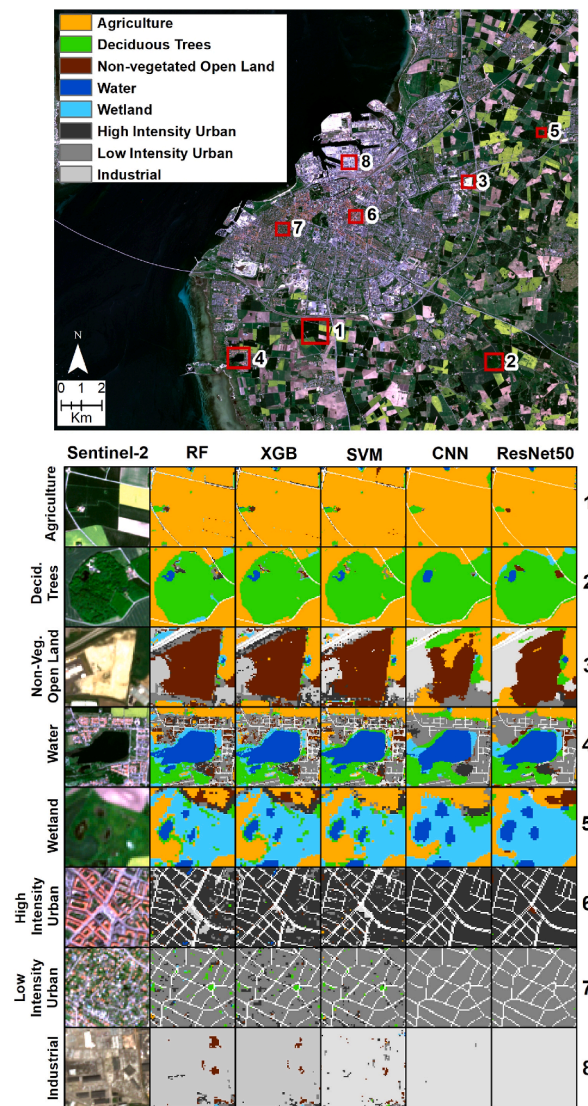


Fig. 5. Side-by-side spatial comparison of the best performing models from the five algorithms with representative examples of each land use and land cover class.

the land surface phenology contributed surprisingly little compared to the raw multispectral data. In particular, it was expected that the NDVI and the start-of-season phenology would be important for the Deciduous Trees and Agriculture classes but their contributions were dwarfed by the near-infrared Sentinel-2 data. Similarly, the GLCM features had negligibly low contributions for all the classes except the High Intensity Urban class.

4. Discussion

4.1. Model comparison

There was a clear pattern of convergence in both precision and recall between the shallow learners (RF, XGB, SVM) and the deep learners (CNN, ResNet50) for classes that were relatively easy to separate such as Agriculture and Water (Fig. 4). The widest disparity between shallow and deep learners was between the classes that were spectrally complex and thereby harder to separate such as Industrial, Low- and High Intensity Urban. The deep learners performed better in these classes with both precision and recall remaining about 85%. This is somewhat expected as deep learners are both based on a 2D CNN architecture, which means that spatial convolutions are applied across the x and y directions (Maggiori et al., 2017) and are invariant to rotation and illumination change (Tong et al., 2018; Yeung et al., 2022). Thus, both deep learners incorporate the spatial information inherent in the satellite data. Since the shallow learners do not explicitly incorporate spatial information, GLCM features were included into the experiments in order to add information about the spatial arrangement of the pixels (Lu and Weng 2007).

It is interesting to note that the GLCM features were part of the best performing model for both RF and XGB (Table 5) but did not feature amongst the most-important variables for any of the classes (Table S6). A probable explanation for this disparity is that the

Table 6

Two-proportion Z-test to compare the proportion of correctly classified pixels for each algorithm and a McNemar's chi-squared (χ^2) test compare the proportions of correctly classified pixels for each class. OA = Overall Accuracy, RF = Random Forest, SVM = Support Vector Machines, XGB = Extreme Gradient Boosting, CNN = Convolutional Neural Network, ResNet50 = Residual Network with 50 layers. Probability values are designated by: *** = $P \leq 0.01$, ** = $P \leq 0.05$, * = $P \leq 0.1$, NS = Not Significant.

Two-Proportion Z-test for OA	Class-wise McNemar's chi-squared test							
	Agriculture	Deciduous Trees	Non-vegetated Open Land	Water	Wetland	High Intensity Urban	Low Intensity Urban	Industrial
RF vs XGB	519.126	0.178	1179.603	4.909	65.675	301.655	170.297	111.637
NS	***	NS	***	**	***	***	***	***
RF vs SVM	326.768	14.871	1231.528	15.393	8.363	556.73	199.114	109.624
NS	***	***	***	***	***	***	***	***
RF vs CNN	359.26	11.442	981.77	0.035	14.37	699.75	202.401	5.285
*	***	***	***	NS	***	***	***	**
RF vs ResNet50	509.008	65.751	645.163	7.961	7.941	228.966	134.497	48.176
**	***	***	***	**	**	***	***	***
XGB vs SVM	143.12	9.27	875.15	1.022	0.346	163.104	42.176	79.71
NS	***	***	***	NS	NS	***	***	***
XGB vs CNN	165.264	6.631	615.51	26.679	0.031	172.513	24.876	0.009
NS	***	***	***	***	NS	***	***	NS
XGB vs ResNet50	272.686	55.672	340.89	70.472	0.614	1.06	3.133	24.685
NS	***	***	***	***	NS	NS	*	**
SVM vs CNN	64.087	47.623	672.498	0.804	33.168	445.425	43.748	0.045
*	***	***	***	NS	***	***	***	NS
SVM vs ResNet50	133.333	141.486	399.133	2.726	44.149	89.817	13.21	24.506
*	***	***	***	*	***	***	***	***
CNN vs. ResNet50	155.127	124.468	158.33	32.476	38.087	100.132	0.735	55.342
NS	***	***	***	***	***	***	NS	***

predictions produced by RF and XGB are averages of several decision trees whereas the additivity property of the SHAP algorithm computes for each class the Shapley value individually for each tree and averages them to produce the value for that class. Since the SHAP algorithm averages the absolute Shapley values for each feature while prioritizing features with large absolute values (Lundberg and Lee 2017), it puts more importance on the magnitude of feature contribution over model performance. Therefore, there is a considerable difference between a variable being important for the overall accuracy of a classifier versus a variable being important for the mapping of individual classes because such a comparison depends on the methodology used to assign importance. If the methodologies are not comparable, e.g. comparing the commonly used permutation feature importance (Altmann et al., 2010) with SHAP feature importance, then any underlying interpretation of this comparison will probably result in spurious conclusions.

The McNemar's chi-squared test (Table 6) showed that 79% of the class-wise pairings were statistically significant at the 1% level ($P \leq 0.01$), suggesting that algorithm accuracies on a class-wise basis are statistically dissimilar. However, the two-proportion Z-test showed that the overall accuracies of statistically similar algorithm pairings had a lower mean difference of correctly classified pixels, while the dissimilar ones had a higher mean difference of correctly classified pixels. This is expected, as the disparity in the PCC between algorithms widens the more dissimilar the predictions (Abdi 2020). An unexpected result is the fact that XGB was the only algorithm where the OA is statistically insignificant with the rest. This was difficult for us to interpret but a possible reason is that XGB represents the midpoint between the lowest (RF = 83.16%) and the highest (ResNet50 = 92.95%) OA values, and the fact that the difference in the PCC between XGB and other algorithms did not exceed the 5400 threshold reported in the results.

All five models provided visually appealing maps (Fig. 5), particularly in the case of the deep learners where the classes were particularly smooth. This is likely related to the aforementioned inclusion of spatial information in the 2D CNN as it can adaptively learn hierarchical spatial structures through backpropagation (Yamashita et al., 2018). However, there may have been some missed opportunities in this regard. For example, in the Low Intensity Urban class the deep learners missed some details such as small-scale urban green spaces such as roundabouts (Fig. 5, row 7). The issue of deep learners oversimplifying complex LULC was also noticed in a crop-land mapping study by Sun et al. (2019) who found that CNNs “ignore pixel-wise temporal patterns in the areas with the complicated context of multiple crop types”. It is difficult to broadly state whether a classifier that produces high OA but ignores small-scale spatial variability (e.g. CNN) outweighs one with slightly lower OA that captures this variation (e.g. XGB) because this depends on the mapping objectives. However, in our case, both the precision and recall of the Low Intensity Urban class were higher for CNN than for XGB (Fig. 4). This indicates that even though the some small-scale spatial variability is not captured by the deep learners, they nonetheless are able to provide an overall better prediction when the class is considered as a whole.

It is clear from our experiments that ResNet50 provided the highest accuracy with the least number of input variables. Thus, one important implication here is that we are able to reduce the computational overhead in large-scale studies, particularly in light of the fact that most users access Earth observation data through download services (Wagemann et al., 2021). Importantly, this allows for the possibility to delve deeper into the biophysical meaning behind the predictions by, for example, closely examining the mechanisms behind the Shapley additive explanations. Ultimately, this would lead to LULC studies that go beyond traditional mapping, and into a more mechanistic understanding of how physical and chemical properties of the urban landscape and its constituent surfaces, as sensed by the satellite, are represented in the algorithm.

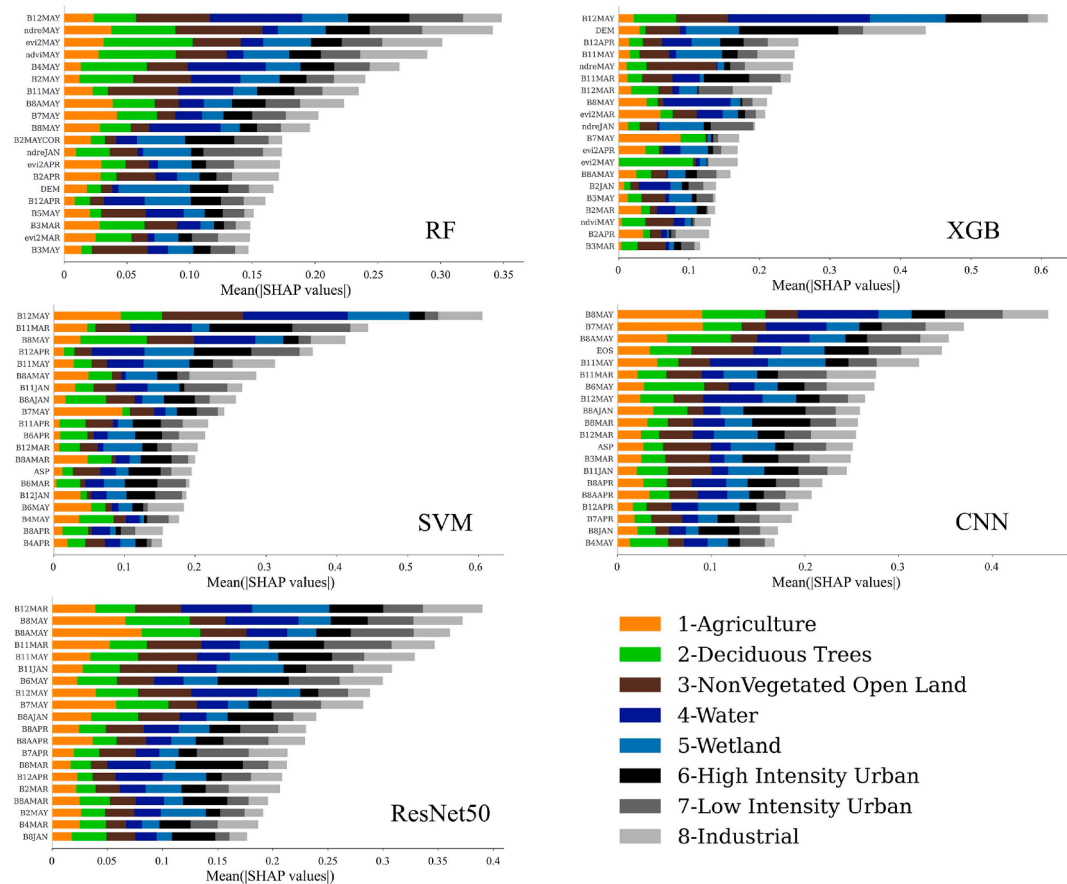


Fig. 6. Shapley values of the explanatory variables showing their contribution on a class-wise basis. For each algorithm, the aggregated Shapley values across all classes per explanatory variable denote the overall importance of that variable.

4.2. Variable importance

Our complete analysis suite involved 141 input features, out of which 14 were the most important in the 85 scenarios we tested (Fig. 7, Supplementary Table S6). The most important variable for each LULC differed between algorithms but they mostly involved Sentinel-2 multispectral features (Table S6). Within these, there were some interesting consistencies. For example, band 7 for the month of May was the most important feature for delineating the Agriculture class for all algorithms except ResNet50 (Fig. 7). Band 7 is a vegetation red edge band, centered at 783 nm, and is responsive to leaf chlorophyll (Sun et al., 2020). A recent study by Mercier et al. (2020) found that band 7, along with band 6, were capable of detecting peak growth in winter cereals and were ranked highly for detecting key phenological stages of the crops. Temporally, (late) May is the month when the main crops of winter barley, winter wheat, and winter rapeseed are in their rapid growth stage as the weather and there is increased irradiance (Wheeler et al., 1996). Another interesting consistency was the fact the most-important features for the Wetland class were mostly the shortwave infrared bands 11 and 12 (centered at 1610 nm and 2190 nm, respectively). The shortwave infrared bands are responsive to vegetation moisture content (Goward 1985) and evidence points to their importance in detecting seasonal variations in plant water status (Wang et al., 2011). Water absorbs highly this portion of the electromagnetic spectrum, so the feasibility of the shortwave infrared band to distinguish wetlands from open water hinges on the presence of a signal from vegetation. Indeed, wetland areas in and around the study area usually have emergent aquatic vegetation (Verchou 2005), and there are often woody species within their vicinity. Comparing the Wetland class to the Water class in Table S6, it is clear that for the latter, the top two variables across all algorithms comprise high water absorption bands (i.e. either near-infrared shortwave infrared). The fact that both shortwave infrared as well as topographic variables are the most important variables for the Wetland class could help distinguish it from the spectrally similar Water class. Although the ability of a DEM to detect wetland microtopography is dependent on its scale and the size of the wetland (Ma et al., 2010), we found DEM features to be useful in delineating wetlands in the study area.

GLCM features were in the best performing model in two of the shallow learners (RF and XGB) (Table 5), however, they did not rank amongst the top twenty most-important variables for those algorithms (Fig. 6). In fact, they rank at the bottom of SHAP hierarchy (Supplementary Figures S3 and S4). This is contrary to other studies that focused on urban landscapes that found improved accuracies when spectral features are supplemented with textural features (Puissant et al., 2005; Shaban and Dikshit 2001). However, Salehi et al. (2012) note that the inclusion of GLCMs does not necessarily improve the classification accuracy and it depends on the

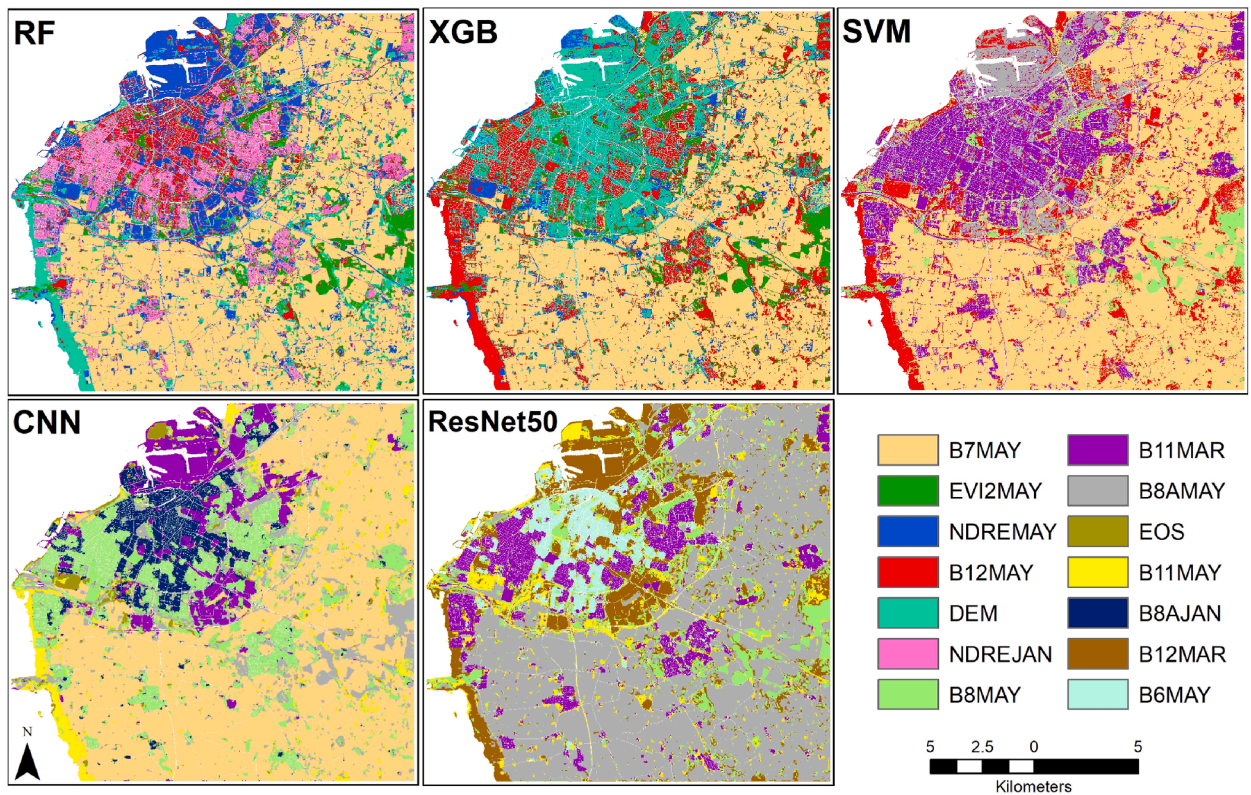


Fig. 7. Spatial distribution of the most-important variables across the study area for each of the mapped classes. Compare these maps with the ones shown in [Supplementary Figure S2](#).

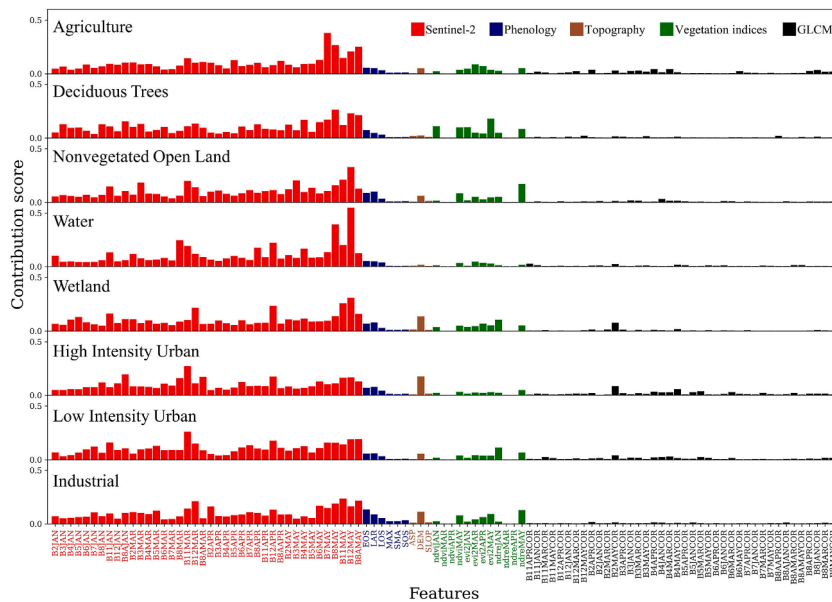


Fig. 8. The Shapley contribution scores for all the explanatory variables grouped by category and land cover and land use class.

specific classes. For example, they report that GLCMs can improve the accuracy of impervious features but not for vegetation and water. A possible reason for the poor performance of GLCMs in our experiments could be that the single-pixel (10 m) distance offset, which was the default in the software, may not have been optimal for the study area because it did not represent the average distance between the urban features in the city. In retrospect, this distance should be calculated through a spatial autocorrelation analysis ([Herold et al., 2003](#)). Another possible reason for the poor performance of the GLCMs is that texture analysis requires higher spatial

resolution than 10 m to be effective in discriminating the structures within the urban fabric (Pesaresi et al., 2016). It is worth nothing, however, that the *Correlation* GLCM of band 2 for the month of May shows a noticeable peak in the Shapley additive explanations (Fig. 8) for both High and Low Intensity Urban areas. Further work is needed to assess the contribution of GLCMs in urban LULC mapping at the native Sentinel-2 resolution.

4.3. Limitations

A limitation of this study is the use of only four Sentinel-2 scenes between the winter and early summer. Although there is evidence that including representative satellite observations across different seasons enhances overall classification accuracy (Chaves et al., 2020), we believe that amount of Sentinel-2 scenes we used was enough to capture the temporal variability of the study area as evidenced by the high overall accuracies (mean = 88%) we obtained. Furthermore, we have incorporated the seasonal progression of the vegetation using the land surface phenology metrics. That said, we may have missed potentially important explanatory variables by not including, for example, scenes from the summer (e.g. June–August) and the fall (e.g. September–October) as these periods have been shown to be important for differentiating trees (Immitzer et al., 2019) and crops (Maponya et al., 2020), respectively. Although we have split the study area into training, test, and validation sub-samples (Table 1), the study area nevertheless largely comprises a single municipal region. This means that decisions on planning, which directly affect LULC, are applied to the entire area. Therefore, a second study area could have aided the interpretation of the Shapley values through either alignment or misalignment with our primary study area. This will be addressed in future research because the current study serves as a proof-of-concept for the application of spatially disaggregated variable importance in LULC mapping.

5. Conclusion

Urban landscapes are generally complex and involve a host of land use and land cover (LULC) stratifications that make mapping them a challenge. Variable importance metrics used in LULC classification assess feature contribution to overall model performance while ignoring the importance of said feature to the individual classes that comprise the final product. The use of importance metrics at the feature level arguably provides more information about the contribution of the explanatory data used to construct the model. This should path the way for more efficient mapping using fewer variables. We showed that not only do overall accuracies increase as one moves from shallow (mean = 84.64%) to deep learners (mean = 92.63%) but that the number of explanatory variables required to obtain maximum accuracy decreases along the same gradient. Furthermore, we demonstrated that class-level importance metrics can be successfully identified using Shapley additive explanations in both shallow and deep learners, which allows for more detailed understanding of variable importance. Out of the 141 input layers tested, 14 stood out as the most important variables in the 17 experiments we performed. These included six spectral bands from May, two spectral bands from March, one spectral band from January, digital elevation model, end of season, and three vegetation indices from May and January. The spectral bands were predominantly red-edge and shortwave infrared, which suggests the sensitivity of these bands to variation in vegetation and the urban fabric. The use of class-level feature importance metrics is still new in LULC mapping, and this study provides important insight into the potential of more nuanced importance metrics. It has to be stressed, however, the purpose of class-level feature importance is not to identify *the single feature* capable of mapping or explaining the variability within a class, but rather the most parsimonious combination of features that can reasonably map a particular class and enhance map accuracy.

Data and code availability

All primary data used in this paper are available for free as open products. The Sentinel-2 satellite data can be downloaded from the Copernicus Open Access Hub (<https://scihub.copernicus.eu>). Geospatial layers from the Swedish Mapping, Cadastral and Land Registration Authority are available from <https://www.lantmateriet.se/en/maps-and-geographic-information>). The Swedish National Land Cover Database can be accessed at <https://www.naturvardsverket.se/verktyg-och-tjanster/kartor-och-karttjanster/nationella-marktackedata>). The European Digital Elevation Model can be downloaded from Copernicus Land Services at <https://land.copernicus.eu/imagery-in-situ/eu-dem>). The machine learning code used to generate the results is available as a Google Colab Notebook at https://colab.research.google.com/drive/1pXvqDzzUFtMbLp1q_U4Zq3YNU_DxzkiF?usp=sharing.

CRedit authorship contribution statement

Benyamin Hosseiny: Formal analysis, Investigation, Validation, Methodology, Visualization, Writing – original draft. **Abdulkhakim M. Abdi:** Conceptualization, Data curation, Project administration, Writing – original draft, Writing – review & editing. **Sadeh Jamali:** Supervision, Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used is publicly available and the code used in the analysis is provided in the manuscript.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.rsase.2022.100843>.

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